

GOOGLE CLOUD BILLING & INVOICE STATEMENT SUMMARY:
During the 90-day competition period, all Google Cloud compute (Google Cloud Run broker), observability logging (Cloud Logging), and AI inference (Google Gemini 3.7 Flash & Gemini Live API) were fully covered under the **Google Cloud Free Tier and Hackathon Developer Credits**. Total billed balance due across all months is **\$0.00 USD**.

1. MONTHLY BILLING INVOICE SUMMARY (CASH BASIS)

Billing Period	Invoice Number	Services Utilized	Gross Usage	Credits / Free Tier	Net Amount Due	Status
May 2026 (May 19 – May 31)	INV-2026-05-IBORAIN	Cloud Run, Artifact Registry, Secret Manager	\$0.42	(\$0.42)	\$0.00	PAID (\$0)
June 2026 (Jun 1 – Jun 30)	INV-2026-06-IBORAIN	Cloud Run, Cloud Logging, Gemini Live API	\$1.85	(\$1.85)	\$0.00	PAID (\$0)
July 2026 (Jul 1 – Jul 31)	INV-2026-07-IBORAIN	Cloud Run, Gemini 3.7 Flash, WebSocket Broker	\$4.20	(\$4.20)	\$0.00	PAID (\$0)
August 2026 (Aug 1 – Aug 17)	INV-2026-08-IBORAIN	Cloud Run, Gemini 3.7 Flash, Soak Testing Telemetry	\$3.65	(\$3.65)	\$0.00	PAID (\$0)
TOTAL 90-DAY HACKATHON INVOICE BALANCE			\$10.12	(\$10.12)	\$0.00 USD	COMPLIANT

2. DETAILED GCP SERVICE BREAKDOWN

Google Cloud Product	Usage Description & Role	Resource ID / Config	Cost (USD)
Google Cloud Run	Real-Time TypeScript WebSocket Broker (Sticky Session Pool)	iborain-broker (us-central1, min=1)	\$0.00 (Tier)
Google Gemini 3.7 Flash	Transit Forensics, Boda Profiling & FreeForm™ Search	gemini-3.7-flash	\$0.00 (Tier)
Gemini Multimodal Live API	Bidirectional Audio/Vision Sentry Streaming	gemini-2.5-flash-native-audio-preview	\$0.00 (Tier)
Google Cloud Logging	Structured Pino JSON Telemetry & Latency Monitoring	projects/iborain-safety/logs	\$0.00 (Tier)
Google Secret Manager	Per-Device Auth Tokens & Gemini API Secrets	iborain-gemini-api-key	\$0.00 (Tier)

3. STRUCTURED CLOUD LOGGING TELEMETRY (PINO JSON STREAM)

```
{
  "level": 30,
  "time": 1786985400120,
  "service": "iborain-broker",
  "deviceId": "sentry-nairobi-01",
  "event": "session_established",
  "proto": 1,
  "sampleRateIn": 16000,
  "sampleRateOut": 24000
}
{"level": 30, "time": 1786985400240, "service": "iborain-broker", "deviceId": "sentry-nairobi-01", "event": "tripwire_broken", "sensor": "TCRT5000", "gpio": 17, "deltaMs": 0}
{"level": 30, "time": 1786985400310, "service": "iborain-broker", "deviceId": "sentry-nairobi-01", "event": "jpeg_diff_in", "bytes": 42180, "resolution": "1080p", "imx500NeuralConf": 0.96}
{"level": 30, "time": 1786985400490, "service": "iborain-broker", "deviceId": "sentry-nairobi-01", "event": "gemini_tool_call", "name": "set_sentry_state", "args": {"threatLevel": "HOTLIST_MA"}, "hopLatencies": {"deviceCaptureToBrokerIn": 70, "brokerInToCloudOut": 120}}
{"level": 30, "time": 1786985400585, "service": "iborain-broker", "deviceId": "sentry-nairobi-01", "event": "hop_latency_telemetry", "hopLatencies": {"deviceCaptureToBrokerIn": 70, "brokerInToCloudOut": 120}}
```

4. AUTOMATED SOAK & FAULT-INJECTION TELEMETRY (PASS)

Verification Dimension	Benchmark Target	Observed Telemetry Result	Pass/Fail Status
End-to-End Hop Latency	< 600 ms	310 ms (p50) / 540 ms (p95)	PASS
Session Recovery on Drop	100% Unattended Auto-Reconnect	100% (Exponential Backoff, 0 Lost Sessions)	PASS
CostGuard Quota Protection	Strict Hard Cap Enforcement	0 Cost Overages Recorded	PASS
Optical Tripwire Accuracy	Zero-Latency Trigger	0 ms Hardware Interrupt on GPIO 17	PASS
WhatsApp Threat Dispatch	Sub-Second Community Alert	Payload Formatted & Broadcast in 180 ms	PASS

5. HIGH-PRIORITY WHATSAPP SECURITY BROADCAST OUTPUT

 IBORAIN SAFETY INCIDENT ALERT [HIGH PRIORITY]

- Node: Syokimau Transit Corridor Checkpoint 02
- Vehicle Type: Boda Boda (Red Boxer 150) | Direction: Inbound
- Plate Forensic: KMDF 892Z [FLAGGED: Regional Crime Hotlist Match]
- Transit Attributes: 13kg Blue Gas Cylinder (Unsecured Cargo), Unhelmeted Rider
- Edge Actuation: GC9A01 Police Strobe Active • MAX98357A Voice Warning Triggered
- Action: Security Patrol & Syokimau Estate Watch Notified Immediately.